

Daily Geometry Drill #2 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations · Tangents · Line–Circle Tests · Chord Lengths.

Section A Rapid Recognition

1. Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.

Answer: "(3, 4)"

Method: Complete the square: $(x - 3)^2 + (y - 4)^2 = 25 - 24 = 1$, so centre (3, 4), radius 1.

Investigate further: In $x^2 + y^2 + 2gx + 2fy + c = 0$ the centre is $(-g, -f)$ and here $2g = -6$, $2f = -8$, so $g = -3$, $f = -4$ — centre (3, 4) ✓. **If the coefficients of x^2, y^2 are not 1, divide through first so they are, then read the centre.**

2. Write down the radius of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.

Answer: 1

Method: From A1, $(x - 3)^2 + (y - 4)^2 = 1$, so $r = \sqrt{1} = 1$.

Investigate further: Formula check: $r = \sqrt{g^2 + f^2 - c} = \sqrt{9 + 16 - 24} = \sqrt{1} = 1$ ✓. **A "circle" with $g^2 + f^2 - c < 0$? Empty locus — the square sum equals a negative number.**

3. Write down the equation of the circle with centre (2, -1) and radius 5.

Answer: " $x^2 + y^2 - 4x + 2y - 20 = 0$ "

Method: $(x - 2)^2 + (y + 1)^2 = 25$ expands to $x^2 + y^2 - 4x + 2y + 5 = 25$, i.e. $x^2 + y^2 - 4x + 2y - 20 = 0$.

Investigate further: Reading back: $2g = -4 \Rightarrow g = -2$ (centre $x = 2$), $2f = 2 \Rightarrow f = 1$ (centre $y = -1$), $r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 1 + 20} = 5$ ✓. **Rewrite for centre (2, -1) radius 2: what becomes of the constant?** $x^2 + y^2 - 4x + 2y + 1 = 0$.

4. Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.

Answer: 5

Method: Complete the square: $(x - 2)^2 + (y + 3)^2 = 4 + 9 + 12 = 25$, so $r = 5$.

Investigate further: Formula: $g = -2, f = 3, r = \sqrt{4 + 9 + 12} = 5$ ✓. **All these squares are perfect: $5^2, 4^2, 3^2$. What radius would an *irrational* answer look like, and would that stop a clean distance test?** Only the appearance changes; the distance test still works.

5. The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x .

Answer: 3

Method: $x^2 + 4^2 = 25 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$; take +3.

Investigate further: Both $(3, 4)$ and $(-3, 4)$ lie on the circle — the $x > 0$ condition kills one branch.

List all lattice points (x, y) **on** $x^2 + y^2 = 25$. $(0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3), (\pm 5, 0)$.

6. The radius from the origin to the point $(3, 4)$ on $x^2 + y^2 = 25$ has what gradient?

Answer: $\frac{4}{3}$

Method: Gradient = $\frac{4-0}{3-0} = \frac{4}{3}$.

Investigate further: The radius is the line joining centre to the point. **Gradient of the perpendicular radius to** $(3, -4)$? $-\frac{4}{3}$.

7. The tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$ has what gradient?

Answer: $-\frac{3}{4}$

Method: Tangent $\perp$ radius, and radius gradient is $\frac{4}{3}$, so tangent gradient = $-\frac{1}{4/3} = -\frac{3}{4}$.

Investigate further: The tangent line is $3x + 4y = 25$ (via $xx_1 + yy_1 = r^2$). **Does it pass through** $(3, 4)$? $9 + 16 = 25$ ✓. **What is its x-intercept?** $x = 25/3$.

8. Write down the equation of the circle with centre $(-1, 2)$ and radius 3.

Answer: " $x^2 + y^2 + 2x - 4y - 4 = 0$ "

Method: $(x + 1)^2 + (y - 2)^2 = 9$ expands to $x^2 + y^2 + 2x - 4y + 5 = 9$, i.e. $x^2 + y^2 + 2x - 4y - 4 = 0$.

Investigate further: Check centre: $g = 1 \Rightarrow$ centre $x = -1$, $f = -2 \Rightarrow$ centre $y = 2$ ✓; $r = \sqrt{1 + 4 + 4} = 3$ ✓. **Where does this circle cross the axes?** Put $y = 0$: $x^2 + 2x - 4 = 0$, a surd — compare with the clean integer axes of Day 1 right-triangles.

9. The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is $(3, 4)$. Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.

Answer: 3

Method: Distance = $\frac{|4(3) + 3(4) - 9|}{\sqrt{4^2 + 3^2}} = \frac{|12 + 12 - 9|}{5} = \frac{15}{5} = 3$.

Investigate further: This is day 1's perpendicular-distance formula applied to a circle's centre. **Since** $d = 3 > r = 1$, **the line misses the circle. How many common tangents could such an external line family have?** Preview of Day 3's tangent-counting.

10. A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Answer: 8

Method: Half-chord = $\sqrt{r^2 - d^2} = \sqrt{25 - 9} = \sqrt{16} = 4$; full length = 8.

Investigate further: The perpendicular from the centre to a chord bisects it — the half-chord

formula. **If the distance were 4, the length is 6; if 5, it degenerates to a tangent. Check both.** The chord shrinks to a point exactly when $d = r$.

Section B Manipulation Drills

1. Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?

A) $x^2 + y^2 - 6x + 4y - 3 = 0$

B) $x^2 + y^2 - 6x + 4y + 9 = 0$

C) $x^2 + y^2 + 6x - 4y - 3 = 0$

D) $x^2 + y^2 - 6x - 4y - 3 = 0$

Answer: A) $x^2 + y^2 - 6x + 4y - 3 = 0$

Method: $(x - 3)^2 + (y + 2)^2 = 16$ gives $x^2 + y^2 - 6x + 4y + 13 = 16$, i.e. $x^2 + y^2 - 6x + 4y - 3 = 0$.

Investigate further: Only A has both the x -coefficient -6 (centre $x = 3$) and the y -coefficient $+4$ (centre $y = -2$). **B has the right coefficients but the wrong constant $+9$; what radius would that give? $r = \sqrt{9 + 4} = 5$.**

2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?

A) 2

B) 3

C) 4

D) $\sqrt{5}$

Answer: C) 4

Method: $(x + 1)^2 + (y - 2)^2 = 1 + 4 + 11 = 16$, so $r = 4$.

Investigate further: Complete the square exactly: $x^2 + 2x = (x + 1)^2 - 1$ and $y^2 - 4y = (y - 2)^2 - 4$, so $(x + 1)^2 + (y - 2)^2 = 11 + 1 + 4 = 16$ and $r = 4$ ✓. With $g = 1, f = -2, c = -11$ the formula reads $r = \sqrt{g^2 + f^2 - c} = \sqrt{1 + 4 + 11}$. **Option D ($\sqrt{5}$) traps students who drop the constant 11 (leaving $r^2 = 1 + 4 = 5$) or flip the sign of c ; keep the sign fixed through the whole square.**

3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?

A) $y = -\frac{1}{2}x + \frac{5}{2}$

B) $y = 2x$

C) $y = -2x + 5$

D) $y = \frac{1}{2}x$

Answer: A) $y = -\frac{1}{2}x + \frac{5}{2}$

Method: Radius gradient $= \frac{2}{1} = 2$; tangent gradient $= -\frac{1}{2}$; through $(1, 2)$: $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$.

Investigate further: Option C uses the wrong sign ($-2x$ is the negative-reciprocal error recalling Day 1), B is the tangent being perpendicular to the perpendicular. **Split-variable test: tangent to $x^2 + y^2 = r^2$ at (x_1, y_1) is $xx_1 + yy_1 = r^2$: here $x + 2y = 5 \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$ ✓.**

4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c .

- A) $c = 1$
 B) $c = 0$
 C) $c = g + f$
 D) $c = -1$

Answer: B) $c = 0$

Method: At the origin $(0, 0) \in$ circle means $0 + 0 + 0 + 0 + c = 0$ — the constant term is the value of the LHS at the origin, so $c = 0$.

Investigate further: Circle $x^2 + y^2 = 25$ has $c = -25$ and indeed does *not* pass through the origin. The constant records the origin's membership: $c = 0 \Leftrightarrow$ origin on the circle.

5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?

- A) $\frac{3}{\sqrt{5}}$
 B) $\frac{3}{5}$
 C) 3
 D) 2

Answer: A) $\frac{3}{\sqrt{5}}$

Method: Line $2x - y + 3 = 0$; distance from $(0, 0) = \frac{|3|}{\sqrt{2^2 + (-1)^2}} = \frac{3}{\sqrt{5}}$.

Investigate further: Compare with $r = 5$: $\frac{3}{\sqrt{5}} < 5$, so the line cuts the circle. This is exactly the Day-1 formula put to the new line-circle test.

6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?

- A) $y = x + 2\sqrt{2}$
 B) $y = x + 4$
 C) $y = 2x + 2$
 D) $y = 2x + 4$

Answer: A) $y = x + 2\sqrt{2}$

Method: Line $x - y + 2\sqrt{2} = 0$ has distance from origin $\frac{2\sqrt{2}}{\sqrt{2}} = 2 = r$ — tangent. The others give $\frac{4}{\sqrt{2}} = 2\sqrt{2}$, $\frac{2}{\sqrt{5}}$, $\frac{4}{\sqrt{5}}$, none equal to 2.

Investigate further: Only option A places the origin exactly r away. **Check B:** $y = x + 4$ has distance $2\sqrt{2} > 2$, so it misses the circle. Shifting a tangent line a little breaks tangency —

| distance must *equal* r .

7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?

- A) 3
B) 2
C) $\sqrt{5}$
D) 1

Answer: D) 1

Method: Half-chord = $2\sqrt{3}$; $(2\sqrt{3})^2 = 13 - d^2 \Rightarrow 12 = 13 - d^2 \Rightarrow d^2 = 1 \Rightarrow d = 1$.

Investigate further: Fallback: remember $(\text{half-chord})^2 + (\perp \text{ dist})^2 = r^2$. Statement C ($\sqrt{5}$) is the trap of $13 - 8$ instead of $13 - 12$.

8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m .

- A) ± 1
B) $\pm\sqrt{3}$
C) ± 2
D) $\pm\sqrt{2}$

Answer: B) $\pm\sqrt{3}$

Method: Line $mx - y + 4 = 0$; tangency requires $\frac{|4|}{\sqrt{m^2+1}} = 2 \Rightarrow \sqrt{m^2+1} = 2 \Rightarrow m^2 = 3 \Rightarrow m = \pm\sqrt{3}$.

Investigate further: Both tangent lines pass through $(0, 4)$, symmetric about the y -axis — the two tangents from an external point. **How long is each tangent segment from $(0, 4)$ to its touch point?** $\sqrt{16 - 4} = \sqrt{12} = 2\sqrt{3}$ — a preview of Power of a Point (Day 5).

9. The circle with centre $(6, a)$ and radius 6 is tangent to the y -axis and passes through the point $(0, 8)$. Find a .

- A) 1
B) 3
C) 5
D) 8

Answer: D) $a = 8$

Method: Tangent to the y -axis ($x = 0$) since centre's x -distance $6 = r$. Through $(0, 8)$: $(0-6)^2 + (8-a)^2 = 36 \Rightarrow 36 + (8-a)^2 = 36 \Rightarrow a = 8$.

Investigate further: Setting $r = 6$ fixes the x -distance; passing through $(0, 8)$ forces the centre level with that point. **Option C (5) assumes the centre is $(6, a)$ with $(8-a) = 1$?** Check $(8-5)^2 = 9 \neq 0$.

10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Answer: $7\sqrt{2}$

Method: Line $x - y + 1 = 0$; distance from the origin $= \frac{|1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$. Half-chord $= \sqrt{25 - \frac{1}{2}} = \sqrt{\frac{49}{2}} = \frac{7}{\sqrt{2}}$; full chord $= \frac{14}{\sqrt{2}} = 7\sqrt{2}$.

Investigate further: Note the line does not pass through the centre, so this is a genuine off-centre chord. If it were $y = x$ the chord would be a full diameter 10.

Section C Substitution & Structure

1. The line $y = 2x + 5$ is a tangent to the circle $x^2 + y^2 = r^2$. Find r^2 .

- A) 1
- B) 5
- C) 25
- D) $5\sqrt{5}$

Answer: B) $r^2 = 5$

Method: Line $2x - y + 5 = 0$; distance from origin $= \frac{5}{\sqrt{4+1}} = \frac{5}{\sqrt{5}} = \sqrt{5} = r$. So $r^2 = 5$.

Investigate further: Option C (25) is “forgot the $\sqrt{m^2 + 1}$ in the denominator”; A is a sign slip.

Tangent to a centred circle has $r = \frac{|c|}{\sqrt{a^2 + b^2}}$.

2. The point $(2, -1)$ lies on the circle $x^2 + y^2 + 2gx + 2fy - 3 = 0$ with $g = f$. Find g .

- A) -1
- B) 0
- C) 1
- D) 2

Answer: A) $g = -1$

Method: Substitute: $4 + 1 + 2g(2) + 2g(-1) - 3 = 0 \Rightarrow 5 + 4g - 2g - 3 = 0 \Rightarrow 2 + 2g = 0 \Rightarrow g = -1$.

Investigate further: Because $g = f$, the two linear terms combine: $2g(x + y)$, and at $(2, -1)$ that is $2g(1) = 2g$. **Checking the centre now:** $g = f = -1$ gives centre $(1, 1)$. The circle passes through $(2, -1)$ but its centre is elsewhere.

3. The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$ have what relationship?

- A) The line cuts the circle in two points (secant)
- B) The line is a tangent
- C) The line misses the circle
- D) The line passes through the centre

Answer: A) secant

Method: Centre $(2, -1)$, $r = \sqrt{4 + 1 + 11} = 4$. Distance to line $x - y + 1 = 0$: $\frac{|2 - (-1) + 1|}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} < 4 = r$ — cuts twice.

Investigate further: Since $2\sqrt{2} < 4$ the line crosses; if = it would be tangent, if > it would miss. Compare the magnitude, never substitute. Option D is the line-through-centre trap: would need $d = 0$.

4. The tangent to $x^2 + y^2 = 9$ at $(2, \sqrt{5})$ has equation?

A) $2x + \sqrt{5}y = 9$

B) $2x - \sqrt{5}y = 9$

C) $2x + \sqrt{5}y = 3$

D) $\sqrt{5}x + 2y = 9$

Answer: A) $2x + \sqrt{5}y = 9$

Method: Split the circle: tangent at (x_1, y_1) is $xx_1 + yy_1 = r^2$, so $2x + \sqrt{5}y = 9$.

Investigate further: Check $(2, \sqrt{5})$ sits on the tangent: $4 + 5 = 9$ ✓, and on the circle: $4 + 5 = 9$ ✓. **Option D swaps the coefficients; does it pass through $(2, \sqrt{5})$?** $\sqrt{5} \cdot 2 + 2\sqrt{5} = 4\sqrt{5} \neq 9$.

5. The midpoint of a chord of $x^2 + y^2 = 40$ is at distance 2 from the centre. Find the chord length.

A) 12

B) $6\sqrt{2}$

C) 8

D) $4\sqrt{10}$

Answer: A) 12

Method: $r = \sqrt{40} = 2\sqrt{10}$; half-chord $= \sqrt{40 - 4} = \sqrt{36} = 6$; chord $= 12$.

Investigate further: **Option D is the “the half-chord is 6 but I forgot to double it” error.** The perpendicular from the centre meets the chord at its midpoint, which is exactly the “distance 2” given.

6. Which of these circles is tangent to the x -axis?

A) $x^2 + y^2 - 6y = 0$

B) $x^2 + y^2 + 4x = 0$

C) $x^2 + y^2 - 6x + 9 = 0$

D) $x^2 + y^2 - 2y - 3 = 0$

Answer: A) $x^2 + y^2 - 6y = 0$

Method: A: centre $(0, 3)$, $r = 3$, distance to x -axis $= 3 = r$ — tangent. B: centre $(-2, 0)$, $r = 2$, distance 0. C: degenerate $r = 0$. D: centre $(0, 1)$, $r = 2$, distance 1.

Investigate further: A circle is tangent to the x -axis exactly when the centre's y -coordinate equals r (up to sign). Option D is a common trap: centre $(0, 1)$, radius 2, so it *crosses* the x -axis rather than touching it.

7. The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ cuts the y -axis at two points. The distance between these two points is?
- A) 4
B) 6
C) 8
D) 10

Answer: D) 10

Method: On the y -axis $x = 0$: $y^2 - 4y - 21 = 0 \Rightarrow (y - 7)(y + 3) = 0$, points $(0, 7)$, $(0, -3)$, distance 10.

Investigate further: Equivalently $d = |\text{product-split}|$; the two intercepts differ by $7 - (-3) = 10$. **This is the chord perpendicular to diameter idea: the y -axis chord here.** Compare the half-chord test: centre $(4, 2)$, $r = \sqrt{16 + 4 + 21} = \sqrt{41}$, distance from centre to y -axis 4, half-chord $\sqrt{41 - 16} = 5$, length 10 ✓.

8. The line $y = kx + 5$ touches the circle $x^2 + y^2 - 6x - 8y + 24 = 0$. How many distinct values of k achieve this?
- A) 0
B) 1
C) 2
D) 3

Answer: C) 2

Method: Circle centre $(3, 4)$, $r = 1$. All lines $y = kx + 5$ pass through $(0, 5)$; distance $OP = \sqrt{(3)^2 + (1)^2} = \sqrt{10} > 1$, so $(0, 5)$ is external: exactly two tangents.

Investigate further: From any external point there are exactly two tangents to a circle — here the two slopes come from $\frac{|3k-4|}{\sqrt{k^2+1}} = 1$. **Option B (1) is tempting if you silently assume $(0, 5)$ is on the circle;** $9 + 1 \neq 1$ so it is not.

Section D Challenge

1. The line $3x + 4y - 12 = 0$ cuts the circle $x^2 + y^2 - 8x - 6y + 16 = 0$ in two points. Find the length of the chord.
- A) $\frac{18}{5}$
B) $\frac{12}{5}$
C) 6
D) $\frac{36}{5}$

Answer: A) $\frac{18}{5}$

Method: Centre $(4, 3)$, $r = \sqrt{16 + 9 - 16} = 3$. Distance to $3x + 4y - 12 = 0$: $\frac{|12+12-12|}{5} = \frac{12}{5}$. Half-chord $= \sqrt{9 - (\frac{12}{5})^2} = \sqrt{\frac{225-144}{25}} = \frac{9}{5}$; chord $= \frac{18}{5}$.

Investigate further: Option B ($\frac{12}{5}$) is the distance-to-centre, not the chord; D doubles it wrongly as $\frac{36}{5}$. **Chain:** distance $\rightarrow$ half-chord (Pythagoras) $\rightarrow$ double. Keep the steps separate.

2. A circle with centre $(3, k)$ is tangent to the x -axis and passes through the point $(0, 4)$. Find the radius.

- A) $\frac{25}{8}$
 B) $\frac{25}{9}$
 C) 3
 D) 4

Answer: A) $\frac{25}{8}$

Method: Tangent to the x -axis $\Rightarrow$ radius $= |k|$. Through $(0, 4)$: $(0-3)^2 + (4-k)^2 = k^2 \Rightarrow 9 + 16 - 8k = 0 \Rightarrow k = \frac{25}{8}$.

Investigate further: The centre moved off the axis, so we cannot read r directly; equate the squared distances. Each candidate option must satisfy both tangent-to-axis and pass-through. Option D (4) is the “ y -coordinate of the point” trap.

3. The tangent to $x^2 + y^2 = 25$ at the point P (in the first quadrant) passes through the point $A = (7, 1)$. Find the x -coordinate of P .

- A) 2
 B) 3
 C) 4
 D) 5

Answer: B) $x = 3$

Method: If $P = (x_1, y_1)$, tangent is $x_1x + y_1y = 25$. Through $A(7, 1)$: $7x_1 + y_1 = 25$; with $x_1^2 + y_1^2 = 25$: $y_1 = 25 - 7x_1 \Rightarrow x_1^2 + (25 - 7x_1)^2 = 25 \Rightarrow 50x_1^2 - 350x_1 + 600 = 0$, factor $(x_1 - 3)(x_1 - 4) = 0$: $x_1 = 3, 4$. Of these, only $x_1 = 3$ gives $y_1 = 4 > 0$ in the first quadrant; $x_1 = 4$ gives $y_1 = -3$.

Investigate further: Two tangent points P satisfy the line-through- A condition; the quadrant chooses one. At $x_1 = 4$, $y_1 = 25 - 28 = -3$, so $(4, -3)$ is in the fourth quadrant.

4. A circle has centre $(1, 4)$ and is tangent to the line $y = 2x + 1$. Find its radius.

- A) $\frac{1}{\sqrt{5}}$
 B) $\frac{1}{5}$
 C) 1

D) $\sqrt{5}$ **Answer:** A) $\frac{1}{\sqrt{5}}$ **Method:** Line $2x - y + 1 = 0$; radius = perpendicular distance from $(1, 4)$: $\frac{|2-4+1|}{\sqrt{4+1}} = \frac{1}{\sqrt{5}}$.**Investigate further:** Option B ($\frac{1}{5}$) ignores the $\sqrt{a^2 + b^2}$; D ($\sqrt{5}$) mis-uses $a^2 + b^2$. **Tangency** $\Leftrightarrow$ **distance from centre = radius — reverse of the line–circle test.**5. The circle through the three points $(0, 0)$, $(6, 0)$ and $(3, -1)$ has what centre?A) $(3, 4)$ B) $(3, 2)$ C) $(4, 3)$ D) $(5, 0)$ **Answer:** A) $(3, 4)$ **Method:** Perpendicular bisector of $(0, 0)$, $(6, 0)$ is $x = 3$. Bisector of $(0, 0)$, $(3, -1)$: midpoint $(1.5, -0.5)$, segment gradient $-\frac{1}{3}$, so bisector gradient 3: $y + 0.5 = 3(x - 1.5) \Rightarrow y = 3x - 5$. At $x = 3$: $y = 9 - 5 = 4$. Centre $(3, 4)$.**Investigate further:** Radius = $\sqrt{9 + 16} = 5$; check all three points are distance 5: to $(6, 0)$ $\sqrt{9 + 16} = 5$ ✓, to $(3, -1) = 5$ ✓. **This is the Day-1 perpendicular-bisector tool finding a circumcentre — the seed of Day 4's triangle centres.**

Top 5 Patterns — Worksheet #2

- Complete the square to read centre and radius.** In $x^2 + y^2 + 2gx + 2fy + c = 0$ the centre is $(-g, -f)$ and $r = \sqrt{g^2 + f^2 - c}$. Strip the linear terms into squares before touching c . (see A1, A2, A4, B2, C8)
- Tangency = distance-to-line equals radius.** For a tangent, $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$. Never substitute line into circle. (see A9, B6, C1, D4)
- Tangent at a point: perpendicular to the radius / split variables.** On $x^2 + y^2 = r^2$, tangent at (x_1, y_1) is $xx_1 + yy_1 = r^2$. (see A7, B3, C4, D3)
- Chord: half-chord = $\sqrt{r^2 - d^2}$.** The perpendicular from the centre bisects the chord and makes a right triangle with half-chord, distance, and radius. (see A10, B7, B10, C5, D1)
- Line–circle positioning by comparing d to r .** $d < r$ secant, $d = r$ tangent, $d > r$ external. (see B5, C3, C8)

Common Traps — Worksheet #2

- Forgetting to halve the linear coefficients.** $x^2 + y^2 - 6x - 8y + 24 = 0$ has centre $(3, 4)$, not $(-6, -8)$. Centre is $(-g, -f)$ where $2g, 2f$ are the linear terms. (see A1, A4, B2)
- Negative reciprocal error on the tangent slope.** Tangent to a circle is perpendicular to the radius, not a mere negation or inversion. (see A6, A7, B3)

- 3. Applying tangency when the point is not on the circle.** The tangent-at-a-point formula only holds for points on the circle; check $x_1^2 + y_1^2 = r^2$ first. *(see B3, C4, D3)*
- 4. Generic lines through the centre are diameters, not tangents.** A line through the centre always cuts twice; tangency needs the perpendicular distance to be exactly r , not 0. *(see B5, C3, C8)*
- 5. Mixing up distance-to-centre with chord length.** The perpendicular distance d is one leg; the half-chord is the other. Squaring and subtracting, then doubling. *(see B7, B10, C5, D1)*

“Tangency is a distance statement in disguise: compare d to r , never solve the intersection.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths